Module – manage installed service's

1. Use the systemctl command to check the status of the sshd service on your Linux system and write down whether it is active, inactive, or failed.

   ANS :

- Command:
  - systemctl status sshd
- Sample Output:
  - sshd.service - OpenSSH server daemon
    Loaded: loaded
    Active: active (running)

- The sshd service is active (running). This means the SSH server is running and accepting remote connections.

2. Start the nginx service using systemctl, then immediately stop it. After each action, verify the status and note the changes.

Hint: Use 'sudo systemctl start nginx' and 'sudo systemctl stop nginx'.

   ANS :

- Start nginx:
  - sudo systemctl start nginx
- Verify Status:
  - systemctl status nginx
- Output:
  - Active: active (running)
- Stop nginx:
  - sudo systemctl stop nginx
- Verify Status:
  - systemctl status nginx
- Output:
  - Active: inactive (dead)

- After starting nginx, the service status changed to active (running). After stopping it, the status changed to inactive (dead).

3. Enable the cron service to start automatically on boot, then reboot your system and confirm that the service is running after reboot.

   ANS :

- Enable Service:
  - sudo systemctl enable crond
- Reboot System:
  - sudo reboot
- Verify After Reboot:
  - systemctl status crond
- Output:
  - Active: active (running)

- The cron service was enabled successfully and started automatically after reboot. The service status is active (running).

4. List all currently running services using systemctl and identify any service related to networking (like NetworkManager or networking). Write the current status of that service.

   ANS :

- Command:
  - systemctl list-units --type=service --state=running
- Sample Networking Service:
  - systemctl status NetworkManager
- Output:
  - Active: active (running)
- The networking service NetworkManager is currently active (running) and is managing network connections.

5. Simulate a real-world scenario: Imagine you are troubleshooting why Spotify Web cannot stream music on your Linux machine. Use systemctl to check if the firewall (ufw or firewalld) service is running, and if it is, disable it temporarily. Document each command used and the result.

Constraint: Only disable the firewall if you are on a safe, non-production environment.

   ANS :

- **For firewalld**
- **Check Status:**
  - systemctl status firewalld

- **Output:**
    - Active: active (running)
- **Stop Firewall Temporarily:**
    - sudo systemctl stop firewalld
- **Verify:**
    - systemctl status firewalld
- **Output:**
    - Active: inactive (dead)

- The firewall service was checked using systemctl. It was found to be running, then temporarily disabled for troubleshooting purposes. After disabling, the service status changed to **inactive**. This should only be done in a safe testing environment and not on a production server.